Danilo Ferreira da **Rocha**

per aspera ad astra



Profile Summary

PhD in Astronomy (Observatório Nacional, 2025), with an internship at the Paris Observatory (2024). I am a Postdoctoral Researcher at the National Astrophysics Laboratory, focusing on massive stars, Be stars, and hierarchical multiple systems. My expertise includes Stellar Astrophysics, Exoplanets, and Astronomy Education and Outreach. Member of the BLOeM collaboration and IOAA 2024 juror.

Contact details

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Personal information

Date of birth: June 10, 1994
Citizenship: Brazilian

Skills

- Linux, Windows, mac-OS
- Fortran, Python, IRAF
- Latex, Microsoft Collection
- Communication/team collaboration

English C1

Useful Links

ORCID: orcid.org/0000-0002-8225-7699
Lattes: lattes.cnpq.br/31704347696753
MyWebsite: danilo-fr.github.io/

Education

Aug 2021 – Oct 2025
National Observatory, Brazil
PhD in Astronomy
Thesis: *Be Stars in Binary Systems*

Aug 2019 – Jul 2021
UERN, Brazil
M.Sc. in Physics

Aug 2013 – Jul 2019
UFAL, Brazil
Degree in Physics

Research Expertise

My research focuses on the evolution of massive stars, with emphasis on evolution, rotation, seismology, binary and hierarchical systems, and the Be phenomenon. I am part of the international BLOeM collaboration, investigating massive-star multiplicity at low metallicity. My work combines multi-wavelength spectroscopy/photometry and time-domain analysis, with interests in exoplanetology.

Selected Appointments/Positions

Nov 2025 – present
LNA, Brazil
Postdoctoral Researcher
Astrophysics research and observational support.

Academic/Professional Activities
2023–present – Reviewer: *The Astrophysical Journal* and A&A;
2014–2026 – Scientific Olympiads Instructor (Physics & Astronomy);
2022–2024 – Student Representative, National Observatory/BR;

Main Publications

*Total: **9** (**2** first-author) peer-reviewed publications · Google Scholar (Aug. 2026: 200 citations, H-index 6)*

The Triple System V1371 Tau: An Eclipsing Binary with an Outer Be Star
D. F. Rocha, M. Emilio, ...
2026 The Astrophysical Journal, 996, 61 DOI: 10.3847/1538-4357/ae1d57

A high fraction of close massive binary stars at low metallicity
H. Sana, T. Shenar, J. Bodensteiner, ..., **D. F. Rocha**, ...
2025 Nature Astronomy, 9, 1337–1346 DOI: 10.1038/s41550-025-02610-x

Observational Experience

PI programs: Gemini North/GMOS (4.5 h, 2026B); SOAR/Goodman (2 nights, 2026A); SOAR/TripleSpec (2 nights, 2024A); TESS General Investigator Program (Cycles 7–9).
Co-I on 10+ additional programs at TESS, VLT, OPD, SOAR, and Gemini.

Honours & Selected Talks

Honours: PhD 10 Award, FAPERJ (2023); Gold Medal, MOBFOG/SAB (2011).
Meeting: Plenary presentations at EANE (2026) and SAB (2023), Cesam2k20 Transport Processes Workshop and VLTI Interferometry School (2024).
Poster: Binary and Multiple Stars in the Era of Big Sky Surveys (2024).

Hobbies

To Travel, Play computer games, Listen to Music and To Cook.

Itajubá-MG/BR, 20th October 2026

Danilo Ferreira da Rocha

Research Projects

My research focuses on the observational and physical processes governing massive stars, from their internal structure to their circumstellar environments, with strong expertise in space- and ground-based instrumentation.

- **Massive stars:** stellar variability and seismology, rotation, spectral modeling, internal mixing processes, and binary/hierarchical stellar evolution.
- **Be phenomenon:** mass ejection mechanisms, formation and dynamics of circumstellar disks, and angular momentum transport.
- **Observational techniques:** multi-wavelength spectroscopy and photometry (IR, optical, UV, X-ray), time-domain analysis, and data-driven modeling.

Academic & Professional Experience

Nov 2025 – present
LNA, Brazil

Postdoctoral Researcher
Astrophysics research and observational support.

Academic Appointments & Teaching

- **2014–2025** Scientific Olympiads Instructor, High School Level.
  Delivered training in physics and astronomy for national and international-level competitions, including OBA and IOAA, fostering student performance and engagement in STEM.
- **2022–2024** Student Representative, Astronomy, National Observatory, Brazil.
- **2014–2019** Teaching Assistant, Physics, Federal University of Alagoas, Brazil.
- **2016–2018** Teaching Assistant, Physics, State Department of Education of Alagoas, Brazil.

Telescope Time

Observing as Principal Investigator

- **4.5 hours** – 8-m Gemini North telescope, GMOS. [GN-2026B-Q-401]
- **2 nights** – 4-m SOAR telescope, Goodman Spectrograph. [SO2026A-018]
- **2 nights** – 4-m SOAR telescope, TripleSpec. [SO2024A-024]
- **TESS General Investigator Program** – Cycles 7 and 9. [G07166, G08147]
- Over 10 additional projects obtained as co-I at TESS, OPD, SOAR, and Gemini telescopes.

Honours and Awards

- **2023** – PhD 10 Award, FAPERJ.
- **2013** – Silver Medal, MOBFOG/SAB and OAFOG.
- **2012** – Honorable Mention, OBMEP/SBM.
- **2011** – Gold Medal, MOBFOG/SAB.

Conferences & Talks

- **2026** – 24th Northeast Astronomy Meeting (EANE), IFAL Satuba, Alagoas, Brazil. Plenary presentation.
- **2025** – 78th SBPC Annual Meeting, UFF, Niterói, Rio de Janeiro, Brazil. LNA panel.
- **2025** – XLVIII Annual Meeting of the Brazilian Astronomical Society, Caxambu, Minas Gerais, Brazil. Parallel presentation.
- **2024** – Cesam2k20 Workshop II: Transport Processes, Observatoire de Paris-Meudon, France.
- **2024** – VLTI Interferometry School, Observatoire de la Côte d'Azur, Porquerolles, France.

- **2023** – XLVI Annual Meeting of the Brazilian Astronomical Society, Rio de Janeiro, Brazil. Plenary presentation.

Posters

- **2024** – Binary and Multiple Stars in the Era of Big Sky Surveys, Czech Republic.

Professional Activities

- **2023–present** – Reviewer, The Astrophysical Journal.
- **2026–present** – Reviewer, A&A.

Event Organization

- **2012–present** – Organizing Committee Member, Alagoas Rocketry Olympiad (Olimpíada Alagoana de Foguetes).
  Responsible for coordination of scientific outreach activities, logistical planning, student engagement, technical guidelines, safety protocols, and teacher training in experimental rocketry.

Open-Source Software Development

WDwrap
Python wrapper for the Wilson–Devinney code, designed to streamline the modeling and parameter inference of binary star systems.

specROTIN-edu
Interactive educational Python tool for demonstrating the effects of stellar rotation on spectral profiles.

Outreach

- **Organizer**, Alagoas Rocketry Olympiad (Olimpíada Alagoana de Foguetes) – Coordinator of a state-level STEM competition, responsible for technical guidelines, safety protocols, and teacher training.
- **Collaborator**, Brazilian Astronomy and Astronautics Olympiad (OBA) – Conducted teacher training workshops and supported regional coordination in northeastern Brazil.
- **Judge**, International Olympiad on Astronomy and Astrophysics (IOAA 2024) – Evaluated theoretical and observational problems as part of the international academic committee.
- **Science Outreach Lecturer** – Delivered public lectures and workshops on stellar astrophysics, exoplanets, and cosmology for high school students and teachers across Brazil.
- **Science Outreach Presenter**, Usina Ciência – Federal University of Alagoas (UFAL).

Media & Public Engagement

- Olhar Digital: Estrelas 'gêmeas'
- Sistemas Binários: Vida e Morte de Estrelas Irmãs / IFBA
- PhD thesis of ON researcher generates article to be published in the Astrophysical Journal / ON

List of Publications

Total: ***9*** *peer-reviewed publications ·* ***2*** *first-author papers* — *Google Scholar (Aug. 2026: 200 citations, H-index 6)*

9 **Binarity at Low Metallicity (BLOeM) – Projected rotational velocities**
D. J. Lennon, S. R. Berlanas, A. Herrero, ..., **D. F. Rocha**, ...
2026 | A&A, 707, A204 | DOI: 10.1051/0004-6361/202558539

8 **The Triple System V1371 Tau: An Eclipsing Binary with an Outer Be Star**
D. F. Rocha, M. Emilio, ...
2026 | The Astrophysical Journal, 996, 61 | DOI: 10.3847/1538-4357/ae1d57

7 **ExoClock Project IV: A homogeneous catalogue of 620 updated exoplanet ephemerides**
A. Kokori, A. Tsiaras, ..., **D. F. Rocha**, ...
2026 | The Astrophysical Journal Supplement Series, 283, 5 | DOI: 10.3847/1538-4365/ae3238

6 **A high fraction of close massive binary stars at low metallicity**
H. Sana, T. Shenar, J. Bodensteiner, ..., **D. F. Rocha**, ...
2025 | Nature Astronomy, 9, 1337–1346 | DOI: 10.1038/s41550-025-02610-x

5 **Binarity at Low Metallicity (BLOeM) – Multiplicity properties of Oe and Be stars**
J. Bodensteiner, T. Shenar, H. Sana, ..., **D. F. Rocha**, ...
2025 | A&A, 698, A38 | DOI: 10.1051/0004-6361/202452623

4 **Binarity at Low Metallicity (BLOeM) – Enhanced multiplicity of early B-type dwarfs and giants at $Z = 0.2Z_{\odot}$**
J. I. Villaseñor, ..., **D. F. Rocha**, ...
2025 | A&A, 698, A41 | DOI: 10.1051/0004-6361/202453166

3 **Binarity at LOw Metallicity (BLOeM) – A spectroscopic VLT monitoring survey of massive stars in the SMC**
T. Shenar, J. Bodensteiner, H. Sana, P. A. Crowther, D. J. Lennon, M. Abdul-Masih, ..., **D. F. Rocha**, ...
2024 | A&A, 690, A289 | DOI: 10.1051/0004-6361/202451586

2 **Distance and age of the massive stellar cluster Westerlund 1. I. Parallax method using Gaia-EDR3**
F. Navarete, A. Damineli, A. E. Ramirez, **D. F. Rocha**, & L. A. Almeida
2022 | MNRAS, 516(1), 1289–1301 | DOI: 10.1093/mnras/stac2374

1 **Distance and age of the massive stellar cluster Westerlund 1–II. The eclipsing binary W36**
D. F. Rocha, L. A. Almeida, A. Damineli, ..., & G. N. Mace
2022 | MNRAS, 517(3), 3749–3762 | DOI: 10.1093/mnras/stac2927